

14/03/19 -

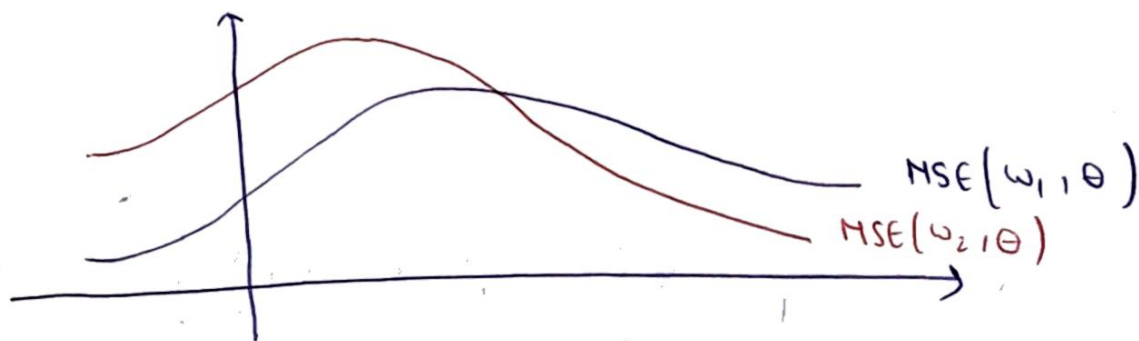
Performance analysis of Estimator -

$$X \equiv (x_1, \dots, x_n) \in \mathbb{R}^n \sim P_\theta, \theta \in \Theta \subseteq \mathbb{R}$$

$$w: \mathbb{R}^n \rightarrow \Theta \subseteq \mathbb{R}$$

$$\begin{aligned} E_\theta \left[(w(x) - \theta)^2 \right] &= \int_{x \sim P_\theta} \left[(w(x) - \theta)^2 \right] \\ &:= \text{MSE}(w, \theta) \end{aligned}$$

$$\left(\begin{array}{l} \theta = \theta_1 \\ \min_w \text{MSE}(w, \theta_1) = 0 \quad \forall \theta_1 \end{array} \right)$$



→ Now if, for $\theta \in \mathbb{R}^d$, $w: \mathbb{R}^n \rightarrow \mathbb{R}^d$

$$\begin{aligned} \text{MSE}(w, \theta) &= E_\theta \left[\sum_{i=1}^d (w(x)_i - \theta_i)^2 \right] \\ &= E_\theta \|w(x) - \theta\|_2^2 \end{aligned}$$

$$\begin{aligned} \rightarrow \text{MSE}(w, \theta) &= E_{\theta} \left[(w(x) - E_{\theta}(w(x))) + (E_{\theta}(w(x)) - \theta) \right]^2 \\ &= E_{\theta} \left[(w(x) - E_{\theta}(w(x)))^2 \right] + E_{\theta} \left[(E_{\theta}(w(x)) - \theta)^2 \right] \\ &\quad + 2 E_{\theta} \left[(w(x) - E_{\theta}(w(x))) (E_{\theta}(w(x)) - \theta) \right] \end{aligned}$$

$$\text{MSE}(w, \theta) = \text{Var}_{\theta}[w] + (\text{Bias}_{\theta}(w))^2 + 0$$

$$\rightarrow \left(\tilde{w}(x) = w(x) + N(0, \sigma^2) \right)$$

- bias remain same
- however variance may increase

e.g. MSE for Normal :

$$X = (X_1, \dots, X_n) \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$$

$\mu, \sigma^2 \equiv \text{unknown } \theta = (\theta_1, \theta_2)$

$$w(X) = \begin{bmatrix} w_1(X) \\ w_2(X) \end{bmatrix}$$

$$\begin{aligned} &\cancel{\mu^2} - \mu^2 - E_{\theta} \left[\left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2 \right] + \\ &\quad \mu E_{\theta} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] \\ &= -\frac{1}{n} \left[\sigma^2 + (n-1)\mu^2 \right] + \mu^2 \end{aligned}$$

e.g. $w_1(x) = \frac{1}{n} \sum_{i=1}^n X_i$

$$(i) E_{\theta}[w_1(x)] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \theta_1$$

Hence $\text{Bias}_{\theta}(w_1) = 0$

$$\begin{aligned} &E_{\theta} \left[\left(\frac{1}{n} \sum_{i=1}^n X_i - \theta_1 \right)^2 \right] \\ &= \frac{1}{n^2} E_{\theta} \left[\sum_{i=1}^n (X_i - \theta_1)^2 \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n E_{\theta} (X_i - \theta_1)^2 \\ &= \frac{1}{n^2} (n\theta_2) \end{aligned}$$

$$(ii) \text{Var}_{\theta}[w_1(x)] = \frac{1}{n^2} n\theta_2 = \frac{\theta_2}{n} = \sigma^2/n$$

$$\text{eg } w_2(x) := \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (56)$$

$$(1) \text{Bias}_\theta(w_2(x)) \Rightarrow$$

$$\begin{aligned} E_\theta[w_2(x)] &= E_\theta \left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right] \\ &= \sum_{i=1}^n E_\theta \left[(x_i - \mu)^2 + (\mu - \bar{x})^2 + 2(x_i - \mu)(\mu - \bar{x}) \right] \end{aligned}$$

$$= n\sigma^2 + nE_\theta(\mu - \bar{x})^2 - n \left(\frac{2}{n} \sigma^2 \right)$$

$$= (n-2)\sigma^2 + n \times \frac{1}{n^2} E_\theta \left[\left(\mu n - \sum_{i=1}^n x_i \right)^2 \right]$$

$$= (n-2)\sigma^2 + \frac{1}{n} E_\theta \left[\left(\sum_{i=1}^n (x_i - \mu) \right)^2 \right]$$

$$= (n-2)\sigma^2 + \frac{1}{n} \sum_{i=1}^n E_\theta (x_i - \mu)^2$$

$$= (n-2)\sigma^2 + \frac{1}{n} n\sigma^2$$

$$= (n-1)\sigma^2$$

therefore,

$$E_\theta[w_2] = \left(\frac{n-1}{n} \right) \sigma^2$$

$$\text{hence } \text{Bias}_\theta(w_2(x)) = \left(\frac{n-1}{n} \right) \sigma^2 - \sigma^2$$

$$\text{Bias}_\theta(w_2(x)) = -\sigma^2/n$$

biased
estimator

$$\begin{aligned} & 2 E_\theta [(x_i - \mu)(\mu - \bar{x})] \\ & 2 E_\theta [\mu x_i - \bar{x} x_i - \mu^2 + \mu \bar{x}] \\ & 2 \left[\mu^2 - E_\theta \left[\left(\frac{1}{n} \sum_{i=1}^n x_i \right) x_i \right] - \mu^2 \right. \\ & \quad \left. + \mu E_\theta \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \right] \\ & 2 \left(-\frac{1}{n} (\sigma^2 + \mu^2) + \mu \frac{1}{n} n\mu \right) \end{aligned}$$

however,

$$\left. \begin{aligned} \tilde{w}_2 &= \frac{1}{(n-1)} \sum_i (x_i - \bar{x})^2 \\ \Rightarrow \text{Bias}_\theta(\tilde{w}_2) &= 0 \end{aligned} \right\}.$$

→ we can check that -

$$\text{Var}_\theta(\tilde{w}_2) = \frac{2\sigma^4}{n-1}$$

$$\Rightarrow \boxed{\text{MSE}(\tilde{w}_2, \theta) = \frac{2\sigma^4}{n-1}}$$

$$\begin{aligned} \text{MSE}(w_2, \theta) &= \left(-\frac{\sigma^2}{n}\right)^2 + \text{Var}_\theta(w_2) \\ &= \left(\frac{\sigma^4}{n^2}\right) + \text{Var}_\theta\left(\tilde{w}_2 \frac{(n-1)}{n}\right) \\ &= \frac{\sigma^4}{n^2} + \frac{(n-1)^2}{n^2} \times \frac{2\sigma^4}{(n-1)} \\ &= \frac{1}{n^2} [\sigma^4 + (n-1)2\sigma^4] \\ &= \frac{1}{n^2} [\cancel{\sigma^4} + 2n\sigma^4 - 2\sigma^4] \\ &= \frac{1}{n^2} [2n-1]\sigma^4 \end{aligned}$$

$$\boxed{\text{MSE}(w_2, \theta) = \frac{(2n-1)}{n^2} \sigma^4}$$

$$\text{Clearly, } \boxed{\text{MSE}(w_2, \theta) < \text{MSE}(\tilde{w}_2, \theta)}$$

→ Hence, it is not necessary to have bias 0 to achieve less / least MSE. } (57)

* Best Unbiased estimator — (BUE)

Defn: An estimator W^* is said to be a best unbiased estimator of function $g(\theta)$ if

(1) $E_{\theta}(W(x)) = g(\theta), \forall \theta \in \Theta$

(2) for any estimator W with $E_{\theta}(W(x)) = g(\theta), \forall \theta \in \Theta$

$Var_{\theta}[W^*(x)] \leq Var_{\theta}[W(x)] \quad \forall \theta \in \Theta$

→ BUE's are also called uniform Minimum Variance unbiased estimators (UMVUE).

eg 2 Best unbiased estimator for Poisson :

$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poi}(d), \quad d \geq 0$

$P_d[X_i = k] = e^{-d} \cdot d^k / k!, \quad k \geq 0$

→ $E_d[X_i] = d = Var_d[X_i]$

(i) $\bar{X} := \frac{1}{n} \sum_{i=1}^n X_i$: has bias = 0

(ii) $\tilde{W}_2(x) := \frac{1}{(n-1)} \sum (X_i - \bar{X})^2$: has bias 0

(iii) $W(x) := \varepsilon \bar{X} + (1-\varepsilon) \tilde{W}_2(x)$: has bias 0

, $\forall 0 \leq \varepsilon \leq 1$

→ we can calculate that

$$\text{Var}_d[\bar{x}] = d/n$$

$$\begin{aligned} \text{Var}_d[\tilde{w}_2(x)] &= d/n + \left(\frac{2n}{n-1}\right) d^2/n \\ &> \text{Var}_d(\bar{x}) \end{aligned} \quad \left. \vphantom{\text{Var}_d[\tilde{w}_2(x)]} \right) \text{clearly}$$

So $\tilde{w}_2(x)$ is ^{worst} ~~Worse~~ than $\bar{x}$ in the MSE sense
but $\bar{x}$ is BLUE?

→ CRAMER-RAO LOWER BOUND
for ESTIMATORS →

(Help us to check that if an unbiased estimator is optimal in MSE. (among all possible unbiased estimators)).

Theorem: (1945-46) let $X \equiv (x_1, \dots, x_n) \in \mathbb{R}$ $\sim P_{\theta}$, $\theta \in \Theta$ and $w(x)$ be any estimator satisfying two conditions

(i) $\text{Var}_{\theta}[w(x)]$ has finite variance $\forall \theta$

$$\underline{\text{Var}_{\theta}[w(x)] < \infty \quad \forall \theta}$$

$$\begin{aligned} \text{(ii)} \quad \frac{d}{d\theta} E_{\theta}[h(x)] &= \frac{d}{d\theta} \int_{\mathbb{R}} h(x) P_{\theta}(x) dx \\ &= \int_{\mathbb{R}} \frac{d}{d\theta} (h(x) P_{\theta}(x)) dx \\ &\quad \forall \theta \in \Theta \end{aligned}$$

$$\left(\text{In general } \frac{d}{d\theta} \int_{\mathbb{R}} u(\theta, x) dx \neq \int_{\mathbb{R}} \frac{d}{d\theta} u(\theta, x) dx \right) \quad (58)$$

→ for $h = w$ and $h = 1$:

Then
 $\forall \theta$

$$\text{Var}_{\theta}[w(x)] \geq \frac{\left(\frac{d}{d\theta} E_{\theta}(w(x)) \right)^2}{E_{\theta} \left[\left(\frac{d}{d\theta} \log p_{\theta}(x) \right)^2 \right]}$$

Usefulness of CRLB:

→ Suppose we want the BLUE for $g(\theta)$,

where $X \sim P_{\theta}$, then if w is unbiased for $g(\theta)$ -

$$E_{\theta}(w(x)) = g(\theta)$$

$$\Rightarrow \text{Var}_{\theta}(w) \geq \frac{(g'(\theta))^2}{E_{\theta} \left[\left(\frac{d}{d\theta} \log p_{\theta}(x) \right)^2 \right]}$$